Dashboard Guide -- Every Section Explained

Intent-Aware Deception System for Real-Time Attacker Profiling · generated from docs/DASHBOARD_GUIDE.md

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Dashboard Guide -- Every Section, Term & Why It Exists

Intent-Aware Deception System for Real-Time Attacker Profiling

> **What this document is.** A complete, screen-by-screen walkthrough of the analyst dashboard: every page, every panel, every chart, every badge, every term -- and *why each one is there*. It is written so you can (a) explain any element in a viva, and (b) paste descriptions straight into your report.

> **Stack:** Vite + React 19 + React Router + Recharts + Tailwind CSS + lucide-react icons. The dashboard is a read-only *analyst console* -- it never sends commands to the honeypot; it only visualises what the backend pipeline has already classified and scored.

> **How data reaches the screen.** Every page fetches from the FastAPI backend over `GET /api/*`. The **Overview** page and the **top navigation bar** additionally hold open two **WebSocket** connections (`/ws/events`, `/ws/alerts`) so new attacks appear live, **without a page refresh**.

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0. Vocabulary -- the core terms first

These terms recur on every screen. Understand these seven and the whole dashboard reads clearly.

Term	Plain meaning	Why the system uses it
Session	One connection from an attacker to the honeypot, from login to disconnect.	The unit we profile -- intent and risk are computed *per session*.
Event	One single action inside a session (one command typed, one login attempt).	The atomic unit the ML model classifies. A session is a sequence of events.
Attacker	A distinct source IP address, aggregated across all its sessions.	Lets us profile a *host* over time, not just one connection.
Intent	The attacker's *goal* behind a command (one of 10 MITRE-aligned classes).	The core research contribution -- we infer *why*, not just *what*.
Attack stage	Where the action sits in the attack lifecycle (kill-chain position).	Turns isolated commands into a *progression* story.
Risk score	A number 0-100 rating how dangerous a session is.	A single, explainable severity signal an analyst can triage on.
Classification source	Which engine decided the intent: `RULE`, `ML`, `HYBRID`, `GEMINI`, or `FALLBACK`.	Transparency -- every decision is auditable, never a black box.

The 10 **intent classes** (used everywhere, always in this order of escalation): `RECONNAISSANCE -> DISCOVERY -> CREDENTIAL_ACCESS -> EXECUTION -> PRIVILEGE_ESCALATION -> PERSISTENCE -> EXPLOITATION -> EXFILTRATION -> CLEANUP -> OTHER`.

The 4 **severity tiers** (derived from the risk score): **LOW** (0-24) - **MEDIUM** (25-49) - **HIGH** (50-74) - **CRITICAL** (75-100, raises an alert).

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1. Global layout (present on every page)

The app shell is: a fixed **top navbar**, a left **sidebar**, and the page content area. Page background `#0a0d14` (near-black navy) -- a dark "security operations centre" theme chosen for long analyst viewing sessions and to make the coloured risk/intent signals pop.

1.1 Top Navbar

- **Left:** a cyan `Shield` icon + the product title **"Intent-Aware Deception System"** and subtitle **"REAL-TIME ATTACKER PROFILING & HONEYPOT TELEMETRY"**.
- **Centre-right:** a live **UTC system clock** (updates every second) -- analysts standardise on UTC so timestamps are unambiguous across time zones.
- **Right: five live status badges.** These are the system's "vital signs", refreshed by polling `GET /api/health` every 15 seconds. Each answers one operational question:

Badge	States (exact text)	What it tells you	Why it matters
Live stream	`LIVE STREAM` / `CONNECTING...` / `DISCONNECTED`	Whether the `/ws/events` WebSocket is connected (green pulsing dot / amber / red)	Confirms you are seeing events in real time, not stale data
Database	`DB: OK` / `DB: ERR`	Whether MongoDB is reachable (`database === 'connected'`)	If the DB is down, nothing persists
ML model	`ML: {N}%` / `ML: ON` / `ML: OFF`	Whether the ML classifier loaded, and its held-out accuracy	Proves the ML core is live; hover shows model name + class count + accuracy
AI / Gemini	`AI: ON` / `AI: OFF`	Whether the optional Gemini layer is configured	Honestly shows the LLM is *optional* -- `OFF` still means fully operational
Honeypot feed	`HONEYPOT: FEED` / `HONEYPOT: IDLE`	Whether the Cowrie log file is present and being read	Confirms the capture source is connected

> **Why these five and not a generic "online" light:** each maps to one independently-failing subsystem (capture -> storage -> ML -> AI -> stream). An analyst can see *exactly* which layer is degraded at a glance. Note the ML and AI badges are deliberately separate -- the design principle "ML is mandatory, Gemini is optional" is made visible right in the header.

1.2 Left Sidebar

Section label **"Security Operations"**, then five navigation links. The active page is highlighted cyan.

Label	Route	Icon	What it opens
Overview	`/`	dashboard grid	The live KPI + charts home page
Live Sessions	`/sessions`	radio	The searchable list of all sessions
Attacker Profiles	`/attackers`	users	Per-source-IP aggregated profiles
Threat Alerts	`/alerts`	alert-octagon	CRITICAL-risk alerts
Analytics	`/analytics`	bar-chart	Cross-dimensional charts

Sidebar footer: a link **"FastAPI OpenAPI Docs"** (opens `/docs` -- the live API explorer) and the caption "Vite + React 19 + Tailwind CSS".

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2. Page -- Overview (`/`)

The home page. Its job: answer *"what is happening right now?"* in one screen. It opens on a row of KPI cards, then telemetry panels. This is the only page (besides the navbar) that consumes **live WebSockets**, so numbers and rows here update on their own.

2.1 KPI stat cards (row of six)

Each is a big number from `GET /api/dashboard/stats`. Visual order:

#	Title	Subtitle (the exact definition shown)	What it counts	Why it's a top-line KPI
1	**Total Sessions**	"Honeypot connections"	All sessions ever captured	Overall engagement volume
2	**Active Sessions**	"Live connections"	Sessions not yet closed	What's happening *now*
3	**Total Attackers**	"Distinct source IPs"	Unique attacker IPs	How many distinct hosts

4	**Total Events**	"Analyzed actions"	Every classified action	Total workload processed (ticks up live)
5	**High Risk**	"Risk 50-74"	Sessions in the HIGH tier	Serious-but-not-critical queue
6	**Critical Alerts**	"Risk >= 75"	Open CRITICAL alerts	The "drop everything" count (ticks up live)

> **Live behaviour:** each `/ws/events` message increments **Total Events**; each `/ws/alerts` message increments **Critical Alerts**. You literally watch the counters move as you attack.

2.2 Panels

Row A 1. **Attack Telemetry Timeline** -- sub *"Events & active sessions volume"*. An area chart of event volume and session volume over time. *Why:* shows bursts of activity -- a spike is a concentrated attack. 2. **Intent Distribution** -- sub *"Classified attacker goals"*. A donut of the 10 intents. *Why:* the at-a-glance answer to "what are attackers mostly trying to do?"

Row B 3. **Threat Severity Tiers** -- sub *"Evaluated session risk levels"*. Bar chart of LOW/MEDIUM/HIGH/CRITICAL counts. *Why:* the shape of your threat queue. 4. **High Risk Alerts** -- sub *"Threshold: Risk Score >= 75"*. The five most recent CRITICAL alerts (`GET /api/alerts?limit=5`). *Why:* immediate triage list. 5. **Live Ingested Telemetry** -- sub *"Streaming real-time actions"*. The eight newest events (`GET /api/events/recent?limit=8`), each with intent, severity, and a **source badge**. *Why:* the live "tail -f" of the honeypot -- this is where you watch each attack land.

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3. Page -- Live Sessions (`/sessions`)

Heading **"Live & Historical Attacker Sessions"**, sub *"Total Captured Sessions: {N}"*. A searchable, paginated table of every session. *Why this page exists:* the Overview shows aggregates; this is where you drill into individual connections.

3.1 Filter toolbar

Filter	Options	API param	Why
Attacker IP	free text (e.g. `192.168.1.100`)	`source_ip`	Find all sessions from one host
Risk Severity	All / LOW / MEDIUM / HIGH / CRITICAL	`severity`	Triage by danger
Dominant Intent	All / the 10 intents	`intent`	Hunt a specific behaviour

Plus **Apply Filter** / **Reset** buttons, a **Refresh** button, and `Page n of m` pagination (15 per page).

3.2 The session table (columns, in order)

Column	Meaning	Why it's shown
Session ID	First 10 chars, links to detail	The identifier
Source IP	Originating attacker IP	Who
Dominant Intent	The session's main goal (colour badge)	The headline classification
Attack Stage	Furthest lifecycle stage reached	How far they got
Risk	Risk score 0-100 with a coloured bar	Danger at a glance
Severity	LOW/MEDIUM/HIGH/CRITICAL badge	Triage tier
Events / Cmds	`total_events / command_count`	Activity volume
Last Seen	Most recent activity time	Recency
Action	**Inspect** link -> session detail	Drill-down

Empty state: *"No Sessions Found -- No attacker sessions matching the query criteria."*

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4. Page -- Session Detail (`/sessions/:id`)

The deepest view: the full **story of one attack**. This is the page that best demonstrates the project's thesis (intent + stage + risk + explainability, per command). It loads three things in parallel: the session, its events (`limit=150`), and its MITRE techniques.

Top bar: a **Back to Sessions** link, and a **Session State** badge -- `ACTIVE` (emerald) or `CLOSED` (slate).

4.1 Hero header card

- **Title** `Session: {id}`; sub-line shows **Source IP** (links to the attacker profile), optional **User**, and **Protocol** (default SSH).
- **Right stat box: Deterministic Threat Risk** (the 0-100 score, large) and **Threat Tier** (severity badge). *"Deterministic"* is deliberate wording -- the risk score comes from a transparent, rule-based engine, not a model guess, so it is reproducible and explainable.
- **Quick metrics (4): Duration** (seconds), **Total Events**, **Commands Executed** (`count (unique)`), **Login Attempts**. *Why:* the behavioural fingerprint of the session -- these same counts feed the risk engine's behavioural boosters.

4.2 Google Gemini AI Session Analyst (purple banner)

- Labelled with an **"ASSISTANT"** pill -- signalling it's an *optional aid*, not the system's decision engine.
- Button **Analyze Session with AI** -> `POST /api/sessions/{id}/analyze`. Produces an **Adversary Assessment Report** with: **Deduced Objective**, **Threat Actor Interpretation**, **Recommended Defensive Response**, and a numbered **Attack Progression Breakdown**. The engine used is shown as a pill (`Engine: ...`).
- *Why it's here and clearly separated:* it turns the structured data into a readable narrative for a report, but the badge/labelling makes clear the core classification already happened without it.

4.3 MITRE + trajectory (two cards)

- **Detected MITRE ATT&CK Techniques** -- one badge per mapped technique (technique ID + name, linking to the official MITRE page). *Why:* grounds every intent in the industry-standard ATT&CK framework -- credibility and shared vocabulary.
- **Multi-Stage Attack Trajectory** -- the **Intent Flow** (chevron-separated sequence of intents in order) and the **Furthest MITRE Stage Reached**. *Why:* this is the "reconnaissance -> discovery -> credential-access -> ..." progression that is the project's headline capability.

4.4 Chronological Attack Timeline & Telemetry Log

The full ordered list of events. Each row shows: index `#`, the **command** (as code) or event type, its **intent badge**, **severity badge**, **stage**, **risk**, local time, and the **explanation** -- either a purple **Gemini AI Analysis** block (`ai_reason`) or a **Rule Match** line (`rule_evidence`). *Why:* every single decision carries its evidence inline -- this is the "explainable" promise made concrete.

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5. Page -- Attacker Profiles (`/attackers`)

Heading **"Attacker Host Intelligence Profiles"**, sub *"Total Unique Attacker IPs: {N}"*. Aggregates *across sessions* by source IP. *Why:* an attacker who returns in five sessions is one threat actor -- this page treats them as one.

5.1 Attacker table (columns, in order)

Column	Meaning
Attacker IP	The host (links to detail)
Total Sessions	How many connections from this IP
Total Events	Total actions across all its sessions
Peak Risk	Highest single-session risk (`maximum_risk`)
Avg Risk	Mean risk across its sessions
Dominant Intent	Its most common goal
Stage	Furthest stage this host reached

Severity	Its worst severity tier
Last Seen	Recency
Action	**Profile** link -> detail

Controls: **Refresh**, an IP **Search** box, and pagination. Empty state: *"No Attacker Profiles -- No distinct attacker hosts identified."*

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6. Page -- Attacker Detail (`/attackers/:ip`)

The per-host dossier. Loads the profile plus that host's sessions (`limit=50`) in parallel.

- **Hero card:** `Attacker IP: {ip}`, **First Seen** / **Last Seen**, and a right stat box with **Peak Risk Score** and **Threat Tier**.
- **Aggregated stats (4): Total Honeypot Sessions**, **Total Ingested Events**, **Average Risk Score**, **Dominant Intent**.
- **Associated Sessions:** heading *"Sessions Originating from {ip}"* then the same session table from §3.2 -- so you can drill from a host straight into any of its sessions.

Why: answers "is this IP a persistent, escalating threat or a one-off scan?" -- exactly the profiling question the project targets.

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7. Page -- Threat Alerts (`/alerts`)

Heading **"Security & Threat Alerts"**, sub *"Trigger condition: Threat Risk Score >= 75 • Total Alerts: {N}"*. *Why a separate page:* CRITICAL sessions are the ones that demand action; they get their own high-visibility, rose-coloured queue.

- **Filters: Status** (All / OPEN / INVESTIGATING / RESOLVED) and **Severity** (All / HIGH / CRITICAL).
- **Alert cards** (not a table -- stacked, red-bordered cards for urgency): each shows the alert ID, **severity** + **intent** badges, a **status** pill, the human-readable **reason**, then a meta line -- **IP**, **Session** (first 10 chars), **Trigger Risk** (`/100`), **Stage**, **Time** -- and an **Inspect Threat Session** button linking straight to §4.
- Empty state (the *good* state): *"Zero Critical Alerts -- No threat sessions have exceeded the critical risk threshold (75/100)."*

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8. Page -- Analytics (`/analytics`)

Heading **"Threat Intelligence Analytics & Security Insights"**, sub *"Cross-dimensional analysis across Honeypot Telemetry, Behavioral Intent, MITRE Attack Stages, and Risk Scores"*. Fetches **six** analytics endpoints in parallel. *Why:* the Overview is "now"; Analytics is the "zoom out and see patterns" page for the report's results section.

Panels, in order:

1. **Attack Telemetry Frequency & Active Sessions Over Time** -- time-series area chart (`/analytics/timeline`).
2. **Intent x Threat Risk Matrix Heatmap** -- a table of intent rows with inline risk meters (`/analytics/heatmap`): per intent, the **event count**, **average risk** (bar + `/100`), **peak risk**, and a low/med/high/crit distribution. *Why:* shows *which intents are most dangerous*, not just most common.
3. **MITRE Attack Stage Distribution** -- horizontal bars of how many sessions reached each lifecycle stage (`/analytics/stages`).
4. **Threat Risk Score Distribution** -- bar chart across the 4 risk tiers (`/analytics/risk-distribution`).
5. **Classified Attacker Intent Distribution** -- the intent donut **plus a breakdown table** (Intent Category - Event Count - Share %).
6. **Session Risk Severity Tiers** -- the severity bar chart **plus a breakdown table** (Severity Tier - Session Count - Share %).

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9. Charts reference (type - what it plots - source)

Chart	Type	Plots	Feed
AttackTimeline	Area (2 series)	Events (cyan) + Sessions (indigo) over time	`/analytics/timeline`
IntentDistributionChart	Donut + 2-column legend	Slice = event count per intent; tooltip shows count + %	`/analytics/intents`

SeverityChart	Vertical bars	Session count per severity tier	`/analytics/severity`
StageDistributionChart	Horizontal bars	Sessions per MITRE stage	`/analytics/stages`
RiskDistributionChart	Vertical bars (4 buckets)	Session count per risk band	`/analytics/risk-distributior`
RiskHeatmap	Table + inline meters	Avg/peak risk + tier split per intent	`/analytics/heatmap`

> The heatmap is intentionally a **table with meters**, not a coloured grid -- it packs four numbers per intent (count, average, peak, distribution) into one readable row, which a pure colour grid can't convey.

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10. Colour & badge legends (why the colours mean what they mean)

Colour is information here, not decoration -- the same intent is always the same colour across every chart and badge, so you learn the palette once.

10.1 Intent colours (all 10)

Intent	Colour	Hex
RECONNAISSANCE	cyan	`#06b6d4`
DISCOVERY	blue	`#3b82f6`
CREDENTIAL_ACCESS	amber	`#f59e0b`
EXECUTION	red	`#ef4444`
PRIVILEGE_ESCALATION	purple	`#a855f7`
PERSISTENCE	indigo	`#6366f1`
EXPLOITATION	rose	`#f43f5e`
EXFILTRATION	fuchsia	`#d946ef`
CLEANUP	teal	`#14b8a6`
OTHER	slate	`#64748b`

Unknown values fall back to OTHER (slate) -- the UI never crashes on an unexpected label.

10.2 Severity tiers

Tier	Risk range	Colour	Note
LOW	0-24	emerald `#10b981`	calm
MEDIUM	25-49	amber `#f59e0b`	watch
HIGH	50-74	orange `#f97316`	serious
CRITICAL	75-100	rose `#f43f5e`	**dot pulses** -- raises an alert

10.3 Risk indicator (the 0-100 bar)

The `{score}/100` indicator recolours by band using the same thresholds: `>=75` rose - `50-74` orange - `25-49` amber - `<25` emerald. Consistent breakpoints mean a colour always implies the same danger, wherever it appears.

10.4 Classification source badges (on every live event)

This is the **transparency layer** -- every event tells you which engine classified it. For `ML`/`HYBRID` it also appends the calibrated confidence (e.g. `ML 92%`).

Source	Colour	Means

HYBRID	cyan	Rule + ML agreed (shows confidence)
ML	violet	ML classifier decided (shows confidence)
RULE	slate	A deterministic rule matched
GEMINI	amber	The optional LLM decided a novel/ambiguous case
FALLBACK	rose	Degraded path (no engine was confident)

> **Why surface the source at all?** Because the project's claim is *explainable* profiling. Seeing `HYBRID 92%` vs `RULE` vs `GEMINI` on each row proves the hybrid engine is real and lets an analyst trust (or question) any single decision.

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11. Live-update mechanics (how "real time" works)

- A reusable `useWebSocket` hook connects over `ws://`/`wss://` (auto-matching the page protocol), reports status `CONNECTING -> LIVE -> DISCONNECTED`, sends a keepalive ping every 15 s, and **auto-reconnects ~3 s** after any drop. *Why:* a dropped socket silently reconnects, so the live view is resilient during a long demo.
- **`/ws/events`** (Overview): new events are de-duplicated by `event_id`, prepended to the live list (kept to 8), bump **Total Events**, and update the intent donut in place.
- **`/ws/alerts`** (Overview): new alerts are de-duplicated by `alert_id`, prepended (kept to 5), and bump **Critical Alerts**.
- Every other page loads on open and via its **Refresh** button / filter changes. *Why the split:* live streaming is valuable on the "now" screen; the analytical pages are snapshots you deliberately refresh.

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12. Complete API surface behind the dashboard

Every screen element above is backed by one of these endpoints (Axios client, base `/api`, 30 s timeout). Useful for the report's "system interface" section.

Endpoint	Feeds
`GET /api/health`	Navbar status badges
`GET /api/dashboard/stats`	Overview KPI cards
`GET /api/events` / `.../recent`	Session detail log / live feed
`GET /api/sessions` / `.../{id}`	Sessions list / detail
`POST /api/sessions/{id}/analyze`	Gemini analyst report
`GET /api/sessions/{id}/mitre`	MITRE techniques card
`GET /api/attackers` / `.../{ip}`	Attacker list / detail
`GET /api/alerts`	Alerts page + Overview alerts panel
`GET /api/analytics/intents`	Intent donut + table
`GET /api/analytics/severity`	Severity bars + table
`GET /api/analytics/timeline`	Timeline area chart
`GET /api/analytics/stages`	Stage distribution bars
`GET /api/analytics/risk-distribution`	Risk band bars
`GET /api/analytics/heatmap`	Intent x risk heatmap
`WS /ws/events`, `WS /ws/alerts`	Live Overview updates

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13. One-paragraph summary (paste-ready)

> The analyst dashboard is a dark-themed React 19 single-page application with five sections -- Overview, Live Sessions, Attacker Profiles, Threat Alerts, and Analytics -- served by a FastAPI backend over REST, with two WebSocket channels driving live updates on the Overview page. It visualises the full profiling pipeline: per-command **intent** (10 MITRE-aligned classes, each with a fixed colour), **attack stage** (kill-chain position), an explainable **risk score** (0-100, four severity tiers, alerting at >=75), and a **classification-source badge** on every event (`RULE`/`ML`/`HYBRID`/`GEMINI`/`FALLBACK`, with ML confidence) that makes each decision auditable. Session-detail and attacker-detail views assemble individual commands into a multi-stage attack **trajectory**, and an *optional* Gemini "Session Analyst" turns that trajectory into a written adversary assessment -- clearly labelled as an assistant, reflecting the system's design principle that ML is mandatory and the LLM is optional.